Indian Institute of Science Education and Research (IISER) Kolkata
Department of Chemical Sciences
CH2201: Fundamentals of Spectroscopy
(Model Solutions)

Mid-Semester Examination **Full Marks: 20** **7 March 2025**

1(a) Rigid symmetry rotor has an energy value equal to $90hB+h(A-B)K^2$. Calculate the highest value of K. **[4]**

(b) For a diatomic molecule the rotational constant obtained from pure rotational spectra is 1.44 cm^{-1}. Calculate the value of rotational quantum number corresponding to the maximum population at 300 K? **[4]**

Solution: 1(a)

J (J +1) = 90 **(1)**
J = -10 and 9 **(1)**
J cannot be –10 and K = + J to –J **(1)**
So the highest value of K = + 9 **(1)**

Solution 1(b)

The rotational quantum number corresponding to the maximum population at 300K is:

$$J_{\max} = \left(\frac{kT}{2Bhc}\right)^{1/2} - \frac{1}{2} \quad \text{when B is in cm}^{-1}$$ **(1)**

$$= \left(\frac{1.38065\times10^{-23}\,JK^{-1}\times300K}{2\times1.44cm^{-1}\times6.626\times10^{-34}\,Js\times3\times10^{10}\,cms^{-1}}\right)^{1/2} - \frac{1}{2}$$

$$= 8$$ **(3)**

2. The wave number of the centre of the fundamental band for CO is 2140 cm^{-1}. Calculate the amplitude of vibration of CO relative to the equilibrium bond length in the ground state when the atoms are at their farthest extremities. [Given: reduced mass of CO = 1.14 x 10^{-23} g, equilibrium bond length of CO = 1.13 Å]. **[8]**

Solution:

The wave number of the centre of the fundamental band for CO is 2140 cm^{-1}.
Hence, its frequency, $\omega = 2140 \times 2.99 \times 10^{10} = 6.4 \times 10^{13}$ per sec. **[1]**
So, force constant, $k = 4\pi^2\omega^2\mu = 4 \times 9.87 \times (6.4 \times 10^{13})^2 \times 1.14 \times 10^{-23} = 18.4 \times 10^5$ dynes/cm **[2]**

Reduced mass of CO, $\mu_{CO} = 1.14. \times 10^{-23}$ g.
At the ground state (v = 0), the vibrational energy,

$$E_0 = \frac{1}{2}\hbar\omega = \left(\frac{1}{2}\frac{h}{2\pi}\sqrt{\frac{k}{\mu}}\right)\frac{1}{2\pi}$$ **[1]**

When the atoms are at their furthest extremities, displacement $x = x_{max}$, the whole of vibrational energy is potential.

So, $E_0 = E_{pot} = \frac{1}{2}kx^2$ **[1]**

$$\frac{1}{2}kx^2 = \left(\frac{1}{2}\frac{h}{2\pi}\sqrt{\frac{k}{\mu}}\right)\frac{1}{2\pi} \qquad \Rightarrow \qquad x_{\max} = \left[\left(\frac{h}{2\pi\sqrt{k\mu}}\right)\frac{1}{2\pi}\right]^{1/2}$$ **[2]**

Putting the corresponding values of k and μ for CO, we have,
$x = 0.02 \times 10^{-8}$ cm
The bond length of CO is 1.13 Å.
Hence, the amplitude is less than 5% of the equilibrium bond distance in the ground state.
[1]

3. If the bond length of H_2 is 0.07417 nm, what will be the positions of the first three Stokes and anti-Stokes Raman lines (in cm^{-1}) in the spectrum relative to the wave number of the incident radiation? [Mass of hydrogen atom = 1.673×10^{-27} kg] **[4]**

Solution:

Reduced mass of the H_2 molecule = $m_H/2 = (1.673 \times 10^{-27})/2$ kg = 8.365×10^{-28} kg **[1]**
The rotational constant,

$$B = \frac{h}{8\pi^2 Ic} = \frac{h}{8\pi^2 \mu r^2 c} = \frac{6.626\times10^{-34}\,Js}{8\times(3.14)^2\times(8.365\times10^{-28}\,kg)\times(0.07417\times10^{-9}\,m)^2\times(3\times10^8\,ms^{-1})}$$

$= 6085\ m^{-1} = 60.85\ cm^{-1}$ **[1]**
So, the first three Stokes lines are at $\bar{\nu} = 6B, 10B \text{ and } 14B$
i.e., 365.1, 608.5 and 851.9 cm^{-1} lower than the incident radiation. **[1]**

The first three anti-Stokes lines are at 365.1, 608.5 and 851.9 cm^{-1} higher than the incident radiation. **[1]**